



AGENT ZERO

Phase 2 Team Checklist — Strengthening Phase

PHASE 2 PLAN

PHASE 2 STRENGTHENING OBJECTIVES

Since the specific hackathon problem statement is not yet announced, Phase 2 focuses on creating an absolute plug-and-play setup. All teammate workloads are divided strictly into separate branches to ensure rapid integration when the competition starts.

TEAM ACTION CHECKLIST

Please complete these tasks inside your respective Git branches. Ensure all API schemas and interfaces strictly match to prevent integration breakage.

Person A — Orchestrator Lead (Owner: branch feat/orchestrator)

feat/orchestrator

☐ Local Docker Orchestration Setup

Write a full docker-compose.yml config to stand up Next.js, Node.js backend, and local pgvector DB in one click.

☐ centralized agents.json Persona Cards

Extract prompt configurations out of javascript into config/agents.json so we can rename and swap agent roles instantly.

☐ LLM Switcher Dashboard Controls

Build dashboard header selectors to dynamically change models (e.g. Gemini-Pro, Claude-Sonnet, Llama) for each agent role.

☐ Exponential Backoff API Retries

Implement safe api request retry wrapper modules to ensure high router reliability when under heavy load.

 Person B — Tools Lead (Owner: branch feat/tools)

feat/tools

☐ **Universal Mocks Feature Switch**

Implement process.env.USE MOCKS config that intercepts all expensive/live tool calls and redirects to mock outputs.

☐ **Skyvern Webhook Callback handler**

Expose a POST endpoint in the server to receive background execution updates from the Skyvern cloud.

☐ **Model Context Protocol (MCP) Client**

Configure a generic MCP loader in tools/ to load local filesystem, shell, and editor MCP server tools.

☐ **N8N Webhook Bridge Tool**

Write a customizable trigger_n8n_workflow tool that delegates long loops or custom webhooks to local/cloud n8n workflows.

 Person C — Memory Lead (Owner: branch feat/memory)

feat/memory

☐ **Reciprocal Rank Fusion (RRF) Hybrid Search**

Rewrite Supabase database retrieval to combine exact keyword matching (FTS) and pgvector semantic retrieval.

☐ **Mem0 Personalization Sync Loop**

Implement a database trigger/loop that pushes updated facts to the user's permanent profile after each step.

☐ **Recursive Knowledge Batch-Indexer**

Provide a console utility to parse, split, and vectorize entire nested document folders automatically.

☐ **Clean Slate Session Restarter**

Build an Express routing endpoint POST /api/memory/reset to clear and recreate memory tables between test runs.